MIT 8.012 Problem Set 4 - Problem 3.7

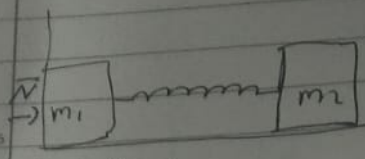
3.7 A system is composed of two blocks of mass m_1 and m_2 connected by a massless spring with spring constant k. The blocks slide on a frictionless plane. The unstretched length of the spring is l. Initially m_2 is held so that the spring is compressed to $l/2$ and m_1 is forced against a stop, as shown. m_2 is released at $t = 0$.



Find the motion of the center of mass of the system as a function of time.

Solution (handwritten) follows on the next pages.

Problem 1 (3.7 K&K)



Notation:-

x_1: pos of m_1 | $\vec{x_1} = x_1 \hat{i}$

x_2: pos of m_2 | $\vec{x_2} = x_2 \hat{i}$

x_3: extension/compression of the spring

$x_3 = -(x_2 - x_1) + \ell$

$= \ell - (x_2 - x_1)$

Δt :- time when block 2 loses contact with the wall

$\vec{R_{cm}}$:- pos of centre of mass

$\vec{V_{cm}}$:- velocity of the centre of mass at $t = \Delta t$

Only external force on the system :- N

$$(m_1 + m_2)\vec{V_{cm}} = \int_0^{\Delta t} N \, dt'$$

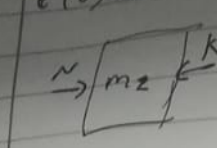
Lemma 1 :- $N = kx_3 \quad \forall t \in (0, \Delta t)$

Pf :-

Assume ~~there does~~

$\exists t$ s.t $N(t) \neq kx_3 \; (\Delta t)$

$\epsilon (0, \Delta t)$

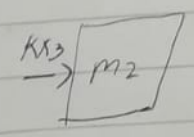


$N - kx_3 \neq 0 \Rightarrow m_2 a \neq 0$

m_2 has already moved and is not in contact with the wall.

But By Assumption $\forall t \in (0, \Delta t)$ m_1 is in contact w/ the wall.

For the sake of contradiction $N = kx_3$



$$\frac{d^2}{dt^2}(x_3) = \frac{d^2}{dt^2}(l - (x_2 - x_1))$$

$$= -a_2 + a_1$$

$$= -a_2 \quad \forall t \in (0, \Delta t)$$

~~$kx_3 = kx_3$~~

$m_2 a_2 = kx_3$

$$\Rightarrow -a_2 = \frac{-kx_3}{m_2} \Rightarrow \frac{d^2(x_3)}{dt^2} = -\frac{kx_3}{m_2}$$

Initial condition $x_3(0) = \frac{l}{2}$

$$\dot{x}_3(t) = \frac{l}{2} \cos\left(\sqrt{\frac{k}{m_2}}\, t\right)$$

Time when block 2 loses contact with the wall is when the spring is stretched ie $x_2 - x_1 > \ell \rightarrow x_3 < 0$

$$\frac{\ell}{2}\cos\left(\sqrt{\frac{K}{m_2}}\,\Delta t\right) = 0 \Rightarrow \Delta t = \sqrt{\frac{m_2}{K}}\,\frac{\pi}{2}$$

$$\overrightarrow{V_{cmi}} = \frac{\ell K}{(m_1+m_2)}\int_0^{\Delta t}\frac{\ell}{2}\cos\left(\sqrt{\frac{K}{m_2}}\,t\right)dt$$

$$= \frac{K}{(m_1+m_2)}\,\frac{\ell}{2}\sqrt{\frac{m_2}{K}}\,\sin\left(\frac{\pi}{2}\right)$$

$$= \frac{\ell\sqrt{Km_2}}{2(m_1+m_2)}$$

Position of the centre of mass for $t > \Delta t$

$$\overrightarrow{R_{cmi}}(t) = R_{cmi}(\Delta t) + \frac{\ell\sqrt{Km_2}}{2(m_1+m_2)}(t-\Delta t)$$

$$= \frac{x_2(\Delta t)m_2}{m_2+m_2} + \frac{\ell}{2}\sqrt{Km_2}\,(t-\Delta t)$$

$$\ell - x_2(\Delta t) = 0 \rightarrow x_2 = \ell$$

$$\overrightarrow{R_{cmi}}(t) = \frac{\ell m_2}{m_2+m_2} + \frac{\ell}{2}\sqrt{Km_2}\,(t-\Delta t)$$